WORKED SOLUTION

STEP II 2005, Question 11

Mechanics • Friction on Inclined Plane

STEP II 2005, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

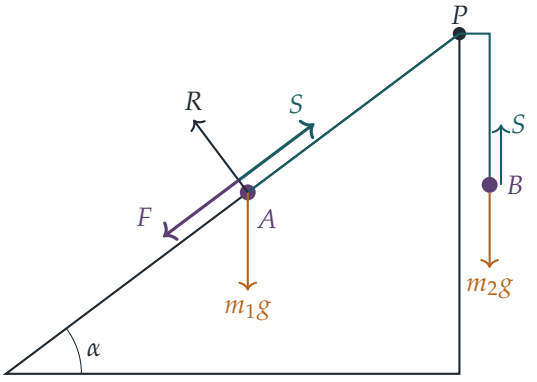
step.maths.org/questions/05-s2-q11

The key idea

With $\tan\alpha = \frac{3}{4}$ we have a 3–4–5 triangle, so $\sin\alpha = \frac{3}{5}$ and $\cos\alpha = \frac{4}{5}$, and every force on A becomes a tidy multiple of $m_1 g$. The motion of A then splits into three stages with constant acceleration: pulled up by the string, sliding up freely, and sliding back down. The direction of friction changes between the last two, and that is where the question hides its subtlety.

Solution

We model A and B as particles, the string as light and inextensible, and the pulley as smooth, so the tension S is the same on both sides. Let $\alpha = \arctan\frac{3}{4}$, so that $\sin\alpha = \frac{3}{5}$ and $\cos\alpha = \frac{4}{5}$.



The acceleration while the string is taut

Perpendicular to the plane, A has no acceleration, so

$$R = m_1 g\cos\alpha = \tfrac{4}{5}m_1 g, \qquad \text{and limiting friction is} \qquad \mu R = \tfrac{1}{2}\cdot\tfrac{4}{5}m_1 g = \tfrac{2}{5}m_1 g.$$

First we check that the system moves. If B were to descend, A would move up the slope, opposed by its weight component $\frac{3}{5}m_1g$ and by friction of at most $\frac{2}{5}m_1g$: a total of at most m_1g. Pulling the other way is B's weight m_2g, and $m_2 > m_1$, so friction cannot hold the system: B descends, A moves up the slope, and friction on A is limiting, acting down the slope.

Let a be the common acceleration. Newton's second law for A (up the slope) and for B (downwards) gives

$$\begin{aligned} m_1a &= S - \tfrac{3}{5}m_1g - \tfrac{2}{5}m_1g = S - m_1g, \\ m_2a &= m_2g - S. \end{aligned}$$

Adding, $(m_1 + m_2)a = (m_2 - m_1)g$, so

$$a = \frac{m_2 - m_1}{m_2 + m_1}\, g,$$

as required. It will be convenient to write $a = kg$, where $k = \dfrac{m_2 - m_1}{m_2 + m_1}$ and $0 < k < 1$.

The time of maximum height

At time T, A is moving up the slope with speed $u = aT = kgT$, having travelled $\frac{1}{2}kgT^2$. When the string breaks, S disappears. A is still moving up the slope, so friction still acts down the slope, and the forces down the slope total $\frac{3}{5}m_1g + \frac{2}{5}m_1g = m_1g$. So A decelerates at exactly g, and comes to rest after a further time

$$\frac{u}{g} = kT.$$

Hence A reaches its maximum height at time

$$T + kT = (1 + k)T = \frac{2m_2}{m_1 + m_2}\, T$$

after release. (We are told A never reaches the pulley, so this stage is not cut short.) During this stage A travels a further $\dfrac{u^2}{2g} = \frac{1}{2}k^2gT^2$, so its total distance up the slope from the point of release is

$$d = \tfrac{1}{2}kgT^2 + \tfrac{1}{2}k^2gT^2 = \tfrac{1}{2}k(1 + k)gT^2.$$

? **Ask yourself** Once A stops, does it stay there?

It stays only if friction can hold it. The weight component down the slope is $\frac{3}{5}m_1g$, while friction can provide at most $\frac{2}{5}m_1g$. Equivalently, $\tan\alpha = \frac{3}{4} > \frac{1}{2} = \mu$. So A slides back down, and now friction acts *up* the slope. The net force down the slope is $\frac{3}{5}m_1g - \frac{2}{5}m_1g = \frac{1}{5}m_1g$, so A accelerates down the slope at $\frac{1}{5}g$, a much gentler rate than its deceleration on the way up.

The ratio of the masses

Starting from rest at its highest point, A takes a time t_d to slide back the distance d to its point of release, where

$$\tfrac{1}{2}\cdot\tfrac{1}{5}g\,t_d^2 = d = \tfrac{1}{2}k(1+k)gT^2, \qquad \text{so} \qquad t_d^2 = 5k(1+k)T^2.$$

We are told that t_d equals the time of maximum height, $(1+k)T$. Squaring, this means

$$(1+k)^2T^2 = 5k(1+k)T^2.$$

Dividing by $(1+k)T^2$, which is positive, gives $1+k=5k$, so $k=\tfrac{1}{4}$. Notice that T has cancelled: this is why the equality holds regardless of when the string breaks, and it pins down k uniquely. Then

$$\frac{m_2-m_1}{m_2+m_1} = \frac{1}{4} \iff 4m_2-4m_1 = m_2+m_1 \iff 3m_2 = 5m_1,$$

so

$$m_1 : m_2 = 3 : 5.$$

Checks

- Limiting cases: if $m_2 \to m_1$ then $k \to 0$ and the time of maximum height tends to T (there is barely any speed to lose); if $m_2 \gg m_1$ then $k \to 1$ and A coasts for as long again as it was pulled, as we'd expect when it reaches speed gT and then decelerates at g.
- With $m_1 = 3$, $m_2 = 5$ and $T = 1\,\text{s}$, we have $k=\tfrac{1}{4}$; maximum height comes at 1.25 s, $d = \tfrac{5}{32}g$, and the descent takes $\sqrt{2d/(g/5)} = \sqrt{25/16} = 1.25\,\text{s}$. A step-by-step numerical simulation of all three stages gave both times as 1.250 s, for several values of T.

Common slips

- Forgetting that friction reverses direction when A starts sliding back, and so using g (or $\tfrac{3}{5}g$) for the downward acceleration instead of $\tfrac{1}{5}g$.
- Measuring the descent only from the point where the string broke, rather than from the point of release.
- Not checking that A actually slides back: this needs $\tan\alpha > \mu$.

Beyond STEP

Friction is the classic example of a force that depends on the state of motion: it switches direction with the velocity and its magnitude is capped while the body is at rest. This makes the motion piecewise, with a different equation of motion in each

stage, and the same idea governs stick–slip phenomena such as squeaking hinges and earthquake faults. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats friction and constrained motion with the care that first-year physics demands.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.